

The Terras coefficient-stopping conjecture and the Pythagorean comma

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July 2026

Abstract

Terras (1976) attached to each positive integer n two quantities: the stopping time $\sigma(n)$, the first iterate of the accelerated Collatz map that falls below n , and the coefficient stopping time $\tau(n)$, the first time the accumulated multiplicative coefficient $3^u/2^t$ falls below 1. Always $\tau(n) \leq \sigma(n)$; the *coefficient stopping time conjecture* (CST) asserts $\tau(n) = \sigma(n)$ for all $n > 1$. We verify CST exhaustively over all 81,119 stopping classes of depth ≤ 24 (covering every n with $\tau(n) \leq 24$; the sole sub-threshold class member is $n = 1$, the trivial cycle, an exact equality case) and directly for all odd $1 < n \leq 10^6$ (zero violations; largest τ observed 176). A violation in the stopping class with u odd steps among j total steps and additive constant b requires $n \leq b/(2^j - 3^u)$. We prove the unconditional bound $b/2^j < u/3$, from which the verified basis yields $\tau(n) = \sigma(n)$ for all $n > 1$ with $\tau(n) \leq 4700$, and, beyond that depth, confines every possible failure of CST to an explicit thin set of classes: those whose (u, j) is a best rational approximation to $\log_2 3$ from above. The smallest safety margins concentrate at $(u, j) = (5, 8)$, the Pythagorean interval $2^8/3^5 = 256/243$, the first nontrivial upper convergent of $\log_2 3$; the next danger zones are $(41, 65)$, $(306, 485)$, and $(15601, 24727)$. The maximal constant $B(t) = \max b/2^j$ over stopping classes of depth t is measured to grow like $1+t/8$, one unit per comma block, with the extremal classes given by repeated comma words; we prove the weaker linear bound $B(t) < t/(3 \log_2 3)$, which suffices for all reductions used. Consequently CST is reduced to effective rational approximation of $\log_2 3$ — the same transcendence wall (Baker, Rhin) that governs the exclusion of nontrivial cycles. One Diophantine question controls both classical subproblems.

1 Introduction

Let T denote the accelerated Collatz (“Terras”) map on the positive integers,

$$T(n) = \begin{cases} n/2, & n \equiv 0 \pmod{2}, \\ (3n+1)/2, & n \equiv 1 \pmod{2}. \end{cases}$$

The $3x+1$ conjecture asserts that every orbit reaches 1; see Lagarias [2] for the classical survey. Terras [1] introduced the *stopping time*

$$\sigma(n) = \inf\{t \geq 1 : T^t(n) < n\},$$

proved that $\sigma(n)$ is finite for almost every n , and observed that the iterate admits the exact affine form

$$T^t(n) = \frac{3^{u_t(n)} n + b_t(n)}{2^t}, \tag{1}$$

where $u_t(n)$ counts the odd iterates among $n, T(n), \dots, T^{t-1}(n)$ and $b_t(n) \geq 0$ depends only on the parity word of the first t steps. The *coefficient stopping time* is

$$\tau(n) = \inf\{t \geq 1 : 3^{u_t(n)} < 2^t\},$$

the first time the multiplicative coefficient in (1) drops below 1. Since $b_t \geq 0$, a drop of the value forces a drop of the coefficient: $\tau(n) \leq \sigma(n)$ for all n . Terras conjectured that the combinatorial quantity captures the arithmetic one exactly:

Conjecture 1.1 (Coefficient stopping time conjecture, CST; Terras 1976). $\tau(n) = \sigma(n)$ for every integer $n > 1$.

The conjecture matters because τ is a function of the residue of n modulo 2^τ alone. If CST holds, the level sets $\{n : \sigma(n) = t\}$ are exactly finite unions of arithmetic progressions modulo 2^t , and all stopping-time densities are exactly computable — this was Terras’ motivation. The exclusion $n > 1$ is forced: $\tau(1) = 2$ while $\sigma(1) = \infty$, because $1 \rightarrow 2 \rightarrow 1$ is the trivial cycle. As we will see, this exception is not an accident but the exact *equality case* of the violation inequality, and the entire conjecture turns out to be governed by how well $\log_2 3 = 1.5849625\dots$ can be approximated from above by rationals j/u — that is, by the continued fraction convergents of $\log_2 3$, whose first nontrivial upper member is the Pythagorean interval $2^8/3^5 = 256/243$.¹

Results

- (a) CST holds for every $n > 1$ with $\tau(n) \leq 24$, by exhaustive symbolic enumeration of all 81,119 stopping classes of depth ≤ 24 (Proposition 3.1). The only class whose minimal member does not clear its violation threshold is the class of $n = 1$, where the threshold is attained with equality — the trivial cycle.
- (b) CST holds for every odd $1 < n \leq 10^6$ by direct computation (zero violations; maximal τ observed 176; Proposition 3.2).
- (c) We prove the unconditional bound $b/2^j < u/3$ for every stopping class (Lemma 5.1). Combined with (b) this gives: CST holds for *all* $n > 1$ with $\tau(n) \leq 4700$ (Corollary 6.2), and in every deeper stopping class except those whose (u, j) lies in an explicit sparse set of upper convergents and semiconvergents of $\log_2 3$ (Theorem 6.1).
- (d) The smallest safety margins concentrate at $(u, j) = (5, 8)$, ratio 256/243; the closest observed approach to a violation is $n = 39$, which after one comma word of eight steps lands on 38, one unit below itself (Section 4).
- (e) The extremal constant $B(t) = \max b/2^j$ over stopping classes of depth t is *measured* to grow linearly, approximately one unit per comma block (1.25, 2.27, 3.24 at $t = 8, 16, 24$), with extremal classes at the repeated comma budgets (5, 8), (10, 16), (15, 24); we give a heuristic sketch of the mechanism (Section 5). Only the proven bound of Lemma 5.1 is used elsewhere in the paper.
- (f) CST is thereby reduced to *effective rational approximation of $\log_2 3$* : lower bounds for $2^j - 3^u$ at the upper convergents, i.e. the territory of Baker’s theorem and Rhin’s effective measures — the identical wall behind the exclusion of nontrivial cycles (Steiner [3], Simons-de Weger [4], Hercher [5]). Section 7 makes the correspondence exact: a cycle attains the violation threshold with equality, so CST violations are precisely *sub-threshold near-cycles*.

Throughout we label every statement as PROVED (full proof supplied or cited), VERIFIED (exhaustive computation, exact integer/rational arithmetic), MEASURED (finite-range computation exhibiting a pattern), or SKETCH (heuristic argument). The one genuinely heuristic item, the sharp B -growth law (e), is quarantined: no other result depends on it.

¹Strictly, 256/243 is the Pythagorean *limma* (the diatonic semitone); the comma proper is $3^{12}/2^{19} = 531441/524288$, attached to the next convergent 19/12, and Mercator’s comma $3^{53}/2^{84}$ belongs to 84/53. We follow loose usage and call all these small intervals “commas.” The from-below members (19/12, 84/53, ...) satisfy $3^u > 2^j$ and are harmless for CST; the danger zones are exactly the from-above members 2/1, 8/5, 65/41, 485/306, 24727/15601, ...

2 Preliminaries

Fix $n \geq 1$ and let $(\epsilon_0, \epsilon_1, \dots)$, $\epsilon_i \in \{0, 1\}$, be the parity word: $\epsilon_i = T^i(n) \bmod 2$. Iterating T gives (1) with $u_t = \epsilon_0 + \dots + \epsilon_{t-1}$ and the constant b_t obeying

$$b_0 = 0, \quad b_{t+1} = \begin{cases} b_t, & \epsilon_t = 0, \\ 3b_t + 2^t, & \epsilon_t = 1. \end{cases} \quad (2)$$

Terras' basic lemma [1] states that the first t parities depend only on $n \bmod 2^t$, and that the map from residues $\bmod 2^t$ to parity words of length t is a bijection. Consequently, for each parity word w of length j there is a unique residue $r \bmod 2^j$ realizing it, and the triple $(u, j, b) = (u_j, j, b_j)$ is a function of w alone.

Definition 2.1 (Stopping class). A parity word w of length j is a *stopping word* if $3^{u_t} \geq 2^t$ for all $t < j$ and $3^{u_j} < 2^j$. The associated *stopping class* is the residue class $\{n \equiv r \pmod{2^j}\}$, recorded as the state (r, u, j, b) . Every $n \geq 1$ with finite τ lies in exactly one stopping class, and $\tau(n) = j$ for all members.

For $j \geq 2$ every stopping word begins with $\epsilon_0 = 1$ (otherwise $3^0 < 2^1$ stops it at $t = 1$), so all members of a stopping class of depth ≥ 2 are odd; conversely all even n satisfy $\tau(n) = \sigma(n) = 1$. Hence CST is a statement about odd n .

Lemma 2.2 (Violation inequality; PROVED). *Let (r, u, j, b) be a stopping class. Every member n satisfies $\sigma(n) \geq \tau(n) = j$, and*

$$\sigma(n) > \tau(n) \iff n \leq \frac{b}{2^j - 3^u} =: \Theta(w).$$

Proof. For $t < j$ we have $3^{u_t} \geq 2^t$ and $b_t \geq 0$, so $T^t(n) = (3^{u_t}n + b_t)/2^t \geq n$: no drop occurs before j , i.e. $\sigma(n) \geq j$. At $t = j$, $T^j(n) < n \iff 3^u n + b < 2^j n \iff n(2^j - 3^u) > b$, and $2^j > 3^u$ by definition of stopping. $\square$

We call $\Theta(w)$ the *violation threshold* of the class and $r/\Theta(w)$ its *margin*: CST holds in the class iff the margin of its least member exceeds 1 (then all members $r + k2^j$ exceed Θ as well).

Lemma 2.3 (The final step is a halving; PROVED). *Every stopping word ends in $\epsilon_{j-1} = 0$, and consequently*

$$j = \lceil u \log_2 3 \rceil.$$

Thus a stopping class' depth is determined by its odd-step count, and

$$1 - \frac{3^u}{2^j} = 1 - 2^{-d(u)}, \quad d(u) := \lceil u \log_2 3 \rceil - u \log_2 3 \in (0, 1).$$

Proof. Suppose $\epsilon_{j-1} = 1$. Then $u = u_{j-1} + 1$ and, since the word did not stop at $j - 1$, $3^{u-1} \geq 2^{j-1}$, whence $3^u \geq 3 \cdot 2^{j-1} > 2^j$, contradicting stopping. So $\epsilon_{j-1} = 0$, $u = u_{j-1}$, and $2^{j-1} \leq 3^u < 2^j$, i.e. $j - 1 \leq u \log_2 3 < j$; since $\log_2 3$ is irrational, $j = \lceil u \log_2 3 \rceil$. $\square$

Lemma 2.3 is the hinge of the whole paper: the denominator $2^j - 3^u$ of the violation threshold is small exactly when $d(u)$ is small, i.e. when $u \log_2 3$ lies just below an integer — when j/u is a good rational approximation of $\log_2 3$ *from above*. By the classical theory of one-sided best approximations (see e.g. Khinchin [8]), such u are precisely the denominators of the upper convergents and upper intermediate fractions (semiconvergents) of the continued fraction

$$\log_2 3 = [1; 1, 1, 2, 2, 3, 1, 5, 2, 23, 2, \dots],$$

with convergents $1, 2, \frac{3}{2}, \frac{8}{5}, \frac{19}{12}, \frac{65}{41}, \frac{84}{53}, \frac{485}{306}, \frac{1054}{665}, \frac{24727}{15601}, \dots$; the upper ones are $\frac{2}{1}, \frac{8}{5}, \frac{65}{41}, \frac{485}{306}, \frac{24727}{15601}, \dots$

3 Exhaustive verification

3.1 Symbolic enumeration of stopping classes

The recursion (2) together with Terras' bijection yields a direct breadth-first enumeration. Maintain the set of *survivor* states (r, u, j, b) (words with $3^{u_t} \geq 2^t$ for all $t \leq j$), starting from $(1, 0, 0, 0)$. Each survivor has two children $r' \in \{r, r + 2^j\}$ modulo 2^{j+1} ; the child's next parity is the parity of $(3^u r' + b)/2^j$, computable by modular arithmetic. A child with $3^{u'} < 2^{j'}$ is a stopping class: record it, compute $\Theta = b'/(2^{j'} - 3^{u'})$ in exact rational arithmetic, and if $r' \leq \Theta$ check the candidate $n = r'$ by direct iteration. All other members of the class exceed r' by multiples of $2^{j'} > \Theta$ in the range explored, so r' is the only candidate.

Proposition 3.1 (VERIFIED). *There are exactly 81,119 stopping classes of depth $j \leq 24$. In exactly one of them the least member fails to exceed the violation threshold: the class $(r, u, j, b) = (1, 1, 2, 1)$ of $n = 1$, where $\Theta = 1/(2^2 - 3^1) = 1$ and the threshold is attained with equality — the trivial cycle $1 \rightarrow 2 \rightarrow 1$. Consequently*

$$\tau(n) = \sigma(n) \quad \text{for every } n > 1 \text{ with } \tau(n) \leq 24.$$

The smallest margin r/Θ among all other classes is 2.020.

(Computation: `scripts/70_cst_attack.py`; integer arithmetic throughout, thresholds as exact fractions; re-run and confirmed for this paper.)

3.2 Direct verification to 10^6

Proposition 3.2 (VERIFIED). *For every odd n with $1 < n \leq 10^6$: $\tau(n) = \sigma(n)$. The maximal coefficient stopping time observed is $\tau = 176$. Together with the trivial even case, CST holds for all $1 < n \leq 10^6$.*

The two propositions overlap but are logically independent: Proposition 3.1 covers all integers of small τ regardless of size; Proposition 3.2 covers all small integers regardless of τ , and it is the second form — a *basis* — that powers the reduction of Section 6, because violation candidates are forced to be small.

4 The margins and the comma

Proposition 3.1 yields margin statistics over all 81,119 classes. The ten smallest margins are:

margin r/Θ	(u, j)	r	$2^j/3^u$
2.020	(5, 8)	39	1.05350
2.391	(3, 5)	11	1.18519
3.805	(17, 27)	251	1.03924
3.965	(5, 8)	79	1.05350
4.200	(2, 4)	3	1.77778
4.507	(4, 7)	7	1.58025
5.013	(5, 8)	123	1.05350
5.853	(5, 8)	95	1.05350
6.053	(3, 5)	23	1.18519
9.141	(5, 8)	199	1.05350

Five of the ten smallest margins, including the minimum, live at $(u, j) = (5, 8)$: the Pythagorean interval

$$\frac{2^8}{3^5} = \frac{256}{243} = 1.053497\dots,$$

the first nontrivial upper convergent $8/5$ of $\log_2 3$. The minimal-margin witness is concrete and pleasing: $n = 39$ has parity word 11101100 ($u = 5$, $j = 8$, $b = 251$, $\Theta = 251/13 = 19.31$), and

$$T^8(39) = \frac{3^5 \cdot 39 + 251}{2^8} = \frac{9728}{256} = 38.$$

After one full comma word, 39 lands one unit below itself. This is the closest approach to a CST violation anywhere in the 81,119 classes.

The mechanism is transparent from Lemmas 2.2 and 2.3: the threshold $\Theta = (b/2^j)/(1 - 3^u/2^j)$ explodes exactly when $1 - 2^{-d(u)}$ is small, i.e. at upper convergents of $\log_2 3$. The relevant data for the first few are (exact rational arithmetic; threshold bound from Lemma 5.1 below):

(u, j)	$2^j/3^u$	$1 - 3^u/2^j$	$\frac{u/3}{1 - 3^u/2^j}$	status
(1, 2)	1.33333	$2.50 \cdot 10^{-1}$	1.34	equality case $n = 1$
(5, 8)	1.05350	$5.08 \cdot 10^{-2}$	32.9	covered ($\geq 2.02\times$)
(41, 65)	1.011529	$1.14 \cdot 10^{-2}$	$1.20 \cdot 10^3$	covered by basis
(306, 485)	1.0010228	$1.02 \cdot 10^{-3}$	$9.99 \cdot 10^4$	covered by basis
(15601, 24727)	1.0000182	$1.82 \cdot 10^{-5}$	$2.86 \cdot 10^8$	open (needs larger basis)

The next danger zone after the comma is thus $(u, j) = (41, 65)$ with $2^{65}/3^{41} = 1.011529\dots$; its threshold bound $1.2 \cdot 10^3$ is comfortably inside the verified basis 10^6 , as is that of (306, 485).² The first zone that escapes the 10^6 basis under the proven bound is the semiconvergent $u = 2966$ (see Corollary 6.2), and the first *convergent* to escape is (15601, 24727).

5 The growth of $B(t)$

The threshold $\Theta = (b/2^j)/(1 - 3^u/2^j)$ has two factors: the Diophantine denominator, discussed above, and the numerator $b/2^j$. Define

$$B(t) = \max\{b/2^j : (r, u, j, b) \text{ a stopping class with } j = t\}.$$

Lemma 5.1 (Linear bound; PROVED). *For every stopping class (r, u, j, b) ,*

$$\frac{b}{2^j} < \frac{u}{3}, \quad \text{hence} \quad \Theta < \frac{u/3}{1 - 3^u/2^j} \quad \text{and} \quad B(t) < \frac{t}{3 \log_2 3} < 0.2104 t.$$

Proof. Set $y_t = b_t/3^{u_t}$, so $y_0 = 0$. By (2), an even step leaves y unchanged and an odd step at depth t gives

$$y_{t+1} = \frac{3b_t + 2^t}{3^{u_t+1}} = y_t + \frac{1}{3} \cdot \frac{2^t}{3^{u_t}} \leq y_t + \frac{1}{3},$$

since the survivor condition at depth t ($3^{u_t} \geq 2^t$) holds for every proper prefix of a stopping word. With u odd steps in total, $y_j \leq u/3$. At the stopping depth, $3^u < 2^j$, so

$$\frac{b}{2^j} = y_j \cdot \frac{3^u}{2^j} < y_j \leq \frac{u}{3}.$$

For the global form, stopping gives $3^u < 2^j$, i.e. $u < j/\log_2 3$. □

This settles, with room to spare, the bounding step that the reduction below requires. The measured truth is sharper and structurally striking:

Observation 5.2 (MEASURED, $t \leq 27$). $B(8) = 1.246$, $B(16) = 2.271$, $B(24) = 3.243$, $B(27) = 4.086$, attained at the classes $(u, j) = (5, 8), (10, 16), (15, 24), (17, 27)$ respectively: the extremal classes carry exactly the repeated comma budget $u/j \approx 5/8$, and B grows by approximately one unit per comma block, $B(t) \approx 1 + t/8$.

²An earlier internal note quoted $2^{65}/3^{41} \approx 1.0035$; the correct value, $36893488147419103232/36472996377170786403 = 1.011529\dots$, is used here.

Remark 5.3 (SKETCH of the mechanism). In the proof of Lemma 5.1 the gain of y at an odd step of depth t is exactly $\frac{1}{3}\rho_t^{-1}$ with $\rho_t = 3^{u_t}/2^t \geq 1$, and the final value is $b/2^j = y_j\rho_j$ with $\rho_j < 1$. Both factors therefore reward trajectories that hug criticality: gains are maximal ($\rightarrow \frac{1}{3}$) when $\rho \approx 1$, and the final discount ρ_j is mildest when the word stops barely below 1. A survivor word that stays near-critical for t steps must spend its odd steps at rate $\approx 1/\log_2 3 = 0.6309$, and the integer word lengths realizing this most efficiently are the comma budgets $(5k, 8k)$ (with the odd steps front-loaded so that every prefix survives). Each 8-step comma block then contributes about $5 \cdot \frac{1}{3} \cdot \langle \rho^{-1} \rangle \approx 1$ to y while ρ decays only by the comma factor $3^5/2^8 = 0.9494$, giving the linear law with slope $\approx 1/8$. We emphasize: this paragraph is a heuristic; the results of Section 6 use only Lemma 5.1.

6 The reduction theorem

Theorem 6.1 (Reduction to the upper convergents; PROVED, with computational input). *Suppose $\tau(n) = \sigma(n)$ has been verified for all $1 < n \leq N$ (as in Proposition 3.2 with $N = 10^6$). Then $\tau(n) = \sigma(n)$ for every $n > 1$ lying in any stopping class (u, j) , $j = \lceil u \log_2 3 \rceil$, with*

$$\frac{u/3}{1 - 2^{-d(u)}} < N, \quad d(u) = \lceil u \log_2 3 \rceil - u \log_2 3.$$

In particular CST can fail only in classes whose (u, j) makes j/u a best one-sided rational approximation of $\log_2 3$ from above — an upper convergent or upper semiconvergent — with $d(u)$ of order u/N or smaller.

Proof. Let $n > 1$ violate CST in such a class. By Lemma 2.2 and Lemma 5.1,

$$n \leq \Theta < \frac{u/3}{1 - 3^u 2^{-j}} = \frac{u/3}{1 - 2^{-d(u)}} < N,$$

using Lemma 2.3 for $j = \lceil u \log_2 3 \rceil$. Classes of depth $j \geq 2$ consist of odd numbers, so n is an odd integer in $(1, N]$, contradicting the verified basis. For the final assertion: if $\frac{u/3}{1 - 2^{-d(u)}} \geq N$ then $1 - 2^{-d(u)} \leq u/(3N)$, so $d(u) \leq \log_2(1 - u/(3N))^{-1} = O(u/N)$; small $d(u)$ forces j/u to approximate $\log_2 3$ from above, and the one-sided best approximations are the upper convergents and semiconvergents [8]. $\square$

Corollary 6.2 (VERIFIED, exact rational arithmetic). *With the basis $N = 10^6$:*

$$\frac{u/3}{1 - 2^{-d(u)}} < 10^6 \quad \text{for every } u \leq 2965,$$

hence $\tau(n) = \sigma(n)$ for every $n > 1$ with $\tau(n) \leq 4700$ (depths $j \leq 4700$ correspond exactly to $u \leq 2965$). The first exceptional odd-step count is $u = 2966$ (a semiconvergent: $2966 = 306 + 4 \cdot 665$), with $j = 4701$ and threshold bound $1.17 \cdot 10^6$, marginally beyond the basis. Among all $u \leq 40,000$ exactly 385 values escape the 10^6 basis, all of them upper semiconvergent or convergent denominators; the largest threshold bound in that range is $2.86 \cdot 10^8$, at $u = 15601$ and $u = 2 \cdot 15601$.

Three remarks on what the reduction does and does not achieve.

(i) *The verified frontier moves from $\tau \leq 24$ to $\tau \leq 4700$.* The enumeration of Proposition 3.1 costs exponential work in the depth; the reduction replaces it, up to depth 4700, by a one-dimensional Diophantine scan plus the fixed basis. Extending the basis to $3 \cdot 10^8$ — a modest computation by current standards (full Collatz convergence has been machine-verified to 2^{68} [9], and a τ - σ check stops at the first drop, on average after ≈ 3.5 steps) — would cover every $u \leq 40,000$, i.e. every $n > 1$ with $\tau(n) \leq 63,397$.

(ii) *No finite basis finishes the job.* Along the upper convergents $d(u) \rightarrow 0$ faster than any fixed multiple of $1/u$ infinitely often (indeed $d(u_k) \asymp 1/(a_{k+1}u_k)$), so the thresholds $\frac{u/3}{1 - 2^{-d(u)}} \asymp u^2 a_{k+1}$ are unbounded: each new upper convergent eventually demands a larger basis. The sequence of demands is, however, only polynomial in the depth if $\log_2 3$ admits an effective irrationality measure, which it does: Baker's theorem [6] gives effective lower bounds for $|j - u \log_2 3|$, and Rhin's Pade-based measure [7] gives, in the form used by Simons and de Weger [4], a bound of the shape $d(u) > c u^{-\kappa}$ with explicit c and κ ($\kappa = 13.3$

in their application). Through Theorem 6.1 this makes the basis requirement for depth T effectively computable and polynomial in T , roughly $O(T^{\kappa+1})$: CST up to any prescribed depth is a *finite, effectively bounded* computation. What no current bound delivers is a uniform closure: full CST would follow from $d(u) \geq cu/N_0$ for a fixed verified basis N_0 and all large u — a strength of effective approximation far beyond present technology.

(iii) *The failure set is thin and explicit.* Any counterexample to CST must be an odd n , larger than every verified basis, sitting inside a specific residue class modulo 2^j whose modulus 2^j is *astronomically* larger than the violation threshold $O(u^2a)$: the class contains at most one candidate below its threshold, namely its least element r , and a violation requires the arithmetically determined r to be freakishly small ($r < \Theta$ while r ranges over $[1, 2^j]$; heuristically probability $\Theta/2^j \approx 0$). CST failures are exactly *near-cycles*, as the next section makes precise.

7 One wall, two sides: CST and cycle exclusion

Suppose n is a periodic point of T with period j and u odd steps per period. Then (1) gives $n = (3^u n + b)/2^j$, i.e.

$$n = \frac{b}{2^j - 3^u} = \Theta(w) : \quad (3)$$

a positive cycle attains the violation threshold of its parity word with equality. The trivial cycle is the instance $(u, j, b) = (1, 2, 1)$, $\Theta = 1$ — precisely the unique sub-threshold member found in Proposition 3.1. Lemma 2.2 says that a CST violation is the inequality relaxation of (3): an integer that does not close the loop but sits at or below the rational fixed point of its own word.³

The known partial results on cycles run through lower bounds on the same denominator $2^j - 3^u$. Steiner [3] excluded nontrivial 1-circuit cycles using Baker’s theorem; Simons and de Weger [4] excluded m -circuit cycles for $m \leq 68$ using Rhin’s effective measure [7]; Hercher [5] pushed this to $m \leq 91$. In every case the engine is: a cycle forces $(2^j - 3^u) \mid b$ with b of size comparable to 2^j , which is incompatible with effective lower bounds for $|2^j - 3^u|$ unless (u, j) is enormous. Theorem 6.1 shows that CST consumes the identical input: effective lower bounds for $2^j - 3^u$ at upper convergents of $\log_2 3$. The two classical subproblems of the $3x + 1$ problem that admit partial rigorous progress — “no nontrivial cycles” and “stopping is decided by the parity word” (CST) — are thus a single question about the effective rational approximation of $\log_2 3$, seen from two sides: equality (divisibility) for cycles, inequality (smallness of the least class member) for CST. The Pythagorean comma $256/243$ is the first nontrivial post of that wall; the margin 2.02 of Section 4 is how close the integers come to it at small depth.

8 Conclusion and honest status

We have verified the Terras coefficient stopping conjecture exhaustively in depth ($\tau \leq 24$, all 81,119 stopping classes) and in height (all $n \leq 10^6$), proved the linear bound $B(t) < t/(3 \log_2 3)$ on the additive constants, and combined the two into an unconditional extension of the verified frontier to $\tau \leq 4700$ together with a complete localization of any possible failure: CST can fail only at explicitly listed upper convergents and semiconvergents of $\log_2 3$, only at integers below an explicitly computable threshold $O(u^2 a_{k+1})$, and only in the guise of a sub-threshold near-cycle. The conjecture is thereby reduced to effective rational approximation of $\log_2 3$ — Baker–Rhin territory — the same wall that bounds cycle exclusion.

Status of each component:

³For a violating n , $T^j(n) \geq n$ with the coefficient already below 1: the orbit has failed to fall despite a falling word — it is shadowing the rational fixed point Θ from below.

Statement	Location	Status
$\tau \leq \sigma$; violation inequality	Lemma 2.2	proved
Final halving; $j = \lceil u \log_2 3 \rceil$	Lemma 2.3	proved
CST for $\tau(n) \leq 24$	Prop. 3.1	verified (exhaustive, exact)
CST for $1 < n \leq 10^6$	Prop. 3.2	verified (exhaustive)
Margin concentration at 256/243	Sec. 4	verified (finite range)
$B(t) < t/(3 \log_2 3)$	Lemma 5.1	proved
$B(t) \approx 1 + t/8$, comma extremizers	Obs. 5.2	measured ($t \leq 27$) + sketch
Reduction theorem	Thm. 6.1	proved (uses Prop. 3.2)
CST for $\tau(n) \leq 4700$	Cor. 6.2	proved + verified scan
Stitching via Rhin-type measures	Sec. 6(ii)	template, not carried out

Two computations would extend the frontier mechanically: (a) a basis to $3 \cdot 10^8$ lifts Corollary 6.2 to $\tau \leq 63,397$; (b) instantiating Rhin’s explicit constants in Theorem 6.1 turns “CST up to depth T ” into an explicit function of feasible computation for every T . The step that would close CST outright — a uniform effective bound $d(u) \gg u/N_0$ — is of the same order of difficulty as excluding all cycles, and we do not claim it. What this note establishes is that both problems now wait at the same door, and that the door is labeled 256/243.

Computational note

All enumerations use exact integer and rational arithmetic (Python, `fractions.Fraction`); the stopping-class enumeration is `scripts/70_cst_attack.py` in the authors’ repository, and every number quoted in this paper (class counts, margins, $B(t)$ values, convergent gaps, the boundary cases $u = 2965, 2966$ of Corollary 6.2) was recomputed independently for the final text.

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